Time Line:

Sunday, February 2, 2025 1:28 PM

Task list:

- AS5600 driver integration
- Test motor + software
- Arduino integration
- Create prototype setup, test balancing
- PCB design
- 3D cad for product
- Test prototype

Date	Task	Deadline	Completed?
11/2/2025	AS5600 driver integration	15/2/2025	
	Test motor + software	22/2/2025	
	Arduino integration	1/3/2025	
	Create prototype setup, test balancing	8/3/2025	
	PCB design	15/3/2025	
	3D cad for product	22/3/2025	
	Test prototype	29/3/2025	

Development Log - AS5600 Encoder Integration

Sunday, February 2, 2025 1:18 PM

1. Encoder Selection Decision
Current Choice: AS5600 (I2C-Based Absolute Encoder)
- Provides absolute position sensing, eliminating the need for homing on startup.
- Seamless startup: offset calibration is needed only once.
- **Limitation:** Update rate is **1.4 ms (~714 Hz max speed update rate)**.

Future Upgrade: AS5048B
- Drop-in replacement with **higher update rate and better performance at high RPMs**.
- Also supports I2C, allowing a smooth transition.
- Maintains absolute position benefits.

2. Communication Protocol Selection
- **I2C chosen** over quadrature signals for AS5600.
- This allows **seamless startup without needing encoder calibration at every boot**.

3. Timer 2 (TIM2) Configuration
TIM2 is used to periodically read AS5600 and compute speed estimates.

Final TIM2 Settings:
- **PSC (Prescaler):** `71`
- **ARR (Auto-Reload Register):** `999`
- **Interrupt Frequency:** `1 kHz (1 ms per tick)`

Rationale:
- **1 ms update rate is stable** and provides sufficient sampling for speed estimation.
- **Future-proofing:** This setting works for **both AS5600 and AS5048B**.
- Avoids excessive CPU load while maintaining responsiveness.

4. Speed Buffering Strategy
- **Reading frequency can be higher than AS5600 update rate (714 Hz)** because:
  - We store **multiple readings in a speed buffer**.
  - The final speed estimate is computed as the **average over the buffer**.
- This ensures smooth speed tracking even if some readings repeat.

5. Future Communication Options
- Since I2C is used for the encoder, **SPI/UART/CAN remains available** for external communication.
- Possible future options for external commands:
  - **UART:** Simple serial communication with another microcontroller.
  - **CAN Bus (if required for robustness).**

Next Steps
- Implement **I2C driver for AS5600** and verify angle readings.
- Develop **speed estimation algorithm using TIM2 interrupts**.
- Validate seamless startup behavior.
- Optimize code structure to support future sensor upgrades.

11/2/2025

Tuesday, February 11, 2025 1:08 PM

Analysis of:
```
__weak void TSK_MediumFrequencyTaskM1(void)
```

What You Need to Modify

Since you are **replacing the encoder driver**, **you need to ensure that**:

1. **ENC_CalcAvrgMecSpeedUnit()** works with the **AS5600 via I2C** instead of the quadrature encoder.
2. **ENC_Clear()** is correctly resetting **your new absolute encoder**.
3. **EAC_StartAlignment()** is modified to **apply the alignment offset to AS5600**.

Modification checklist:

Function	Modified?
`(void)ENC_CalcAngle(&ENCODER_M1);`	
ENC_CalcAvrgMecSpeedUnit()	
ENC_Clear()	
EAC_StartAlignment()	

`__weak void *ENC_IRQHandler(void *pHandleVoid)`
`->void SPD_TIM_M1_IRQHandler(void)`
 	
AS5600.c -> mc_common_it.c	

20/2/2025

Thursday, February 20, 2025 11:25 PM

I need to find a better way to be consistent… its like I take a week break between sessions each time…

I have one month before the exam study period. Ideally I already have a working driver, code and algorithm. Then only PCB design and CAD is necessary (can be done after exams)

Tasks completed:

- Fixed tim2: from no output compare -> internal clock
    - Also removed independent irq handler and now uses integrated: `*ENC_IRQHandler`
- ENC init and ENC Clear are ready

Next step:

- Check ENC_CalcAngle for correctness

21/02/2025

Friday, February 21, 2025 8:16 PM

Made good progress yesterday so I was motivated to work today aswell

22/02/2025

Saturday, February 22, 2025 9:44 PM

Another day another dollar. Lets work on the alignment control script?
Also I have a feeling that tim2 is not necessary at all.. It was previously used to count encoder input pulses… now im trying to use it to force a sampling frequency but there should already be an timer that handles the sampling frequency… lets investigate..

23/02/2025

Sunday, February 23, 2025 10:57 PM

Yesterday I completed driver code but (in theory)
Now I need to do actual testing…

```
make -j22 all" terminated with exit code 2. Build might be inco

23:15:02 Build Failed. 154 errors, 0 warnings. (took 3s.955ms)
```

Excellent….. 153 errors!!
Lol

24/02/2025

Monday, February 24, 2025 11:41 PM

Lets see if the driver I wrote is correct…

No progress today. Hit a wall with the debugging. Prob due to motor control init not interfacing well with my driver… need to investigate somehow…

25/02/2025

Tuesday, February 25, 2025 4:24 PM

My work streak is fading did almost no work yesterday and not motivated to work today… nevertheless I should investigate further why its not working…

What motivates me:

- Idea that I am 1 step away from having a working driver
- I fixed my motorbike on my own by welding, blowtorching, and elbow greasing my way through it, so debugging this little program should be no problem!

Busy debugging today:

- Need to figure out how to console.log things to check variables and if code is running
- Need to ensure that motorcontrol tasks is being called by systick in mc_common_it
    - Which then calls med freq task which then allows the statemachine to work
    - I know that its trying to toggle from idle, but the change command is not being acknowledged..